

Polynomial Regression

Assignment 1 | BT2024260 | 8 October 2026

Two independent polynomial ridge models were trained to predict Net Power Score (var1) and Thermal Anomaly Score (var2). Degree and regularization were selected by three-fold cross-validation on development data. A separate grouped holdout was then evaluated, before each final model was refitted on all labeled data.

1. Problems and data

Dataset	Inputs and target	Train / test rows
var1	Six plant settings, x1-x6; Net Power Score y	1,000 / 1,000
var2	Three coordinate offsets, x1-x3; Thermal Anomaly Score y	1,000 / 1,000

Only the four files matching roll number BT2024260 were used. All input and target values were finite, with no missing entries. Input values in both problems lie in $[-1, 1]$. No imputation, additional input scaling, or target transformation was needed. The supplied sample submission contains one column, **y**, and 1,000 rows.

Repeated input vectors occur in the training sets: one extra repeated row in var1 and nine in var2. Their responses were retained. Identical input vectors were kept together in every split to prevent a repeated location or setting from appearing on both sides of a validation split.

2. Selected models and holdout results

Problem	Degree	Terms*	Ridge alpha	MSE	R2
var1	5	461	3.162278	0.497835	0.961435
var2	13	559	0.316228	0.252532	0.994440

*Nonconstant polynomial terms; a separate intercept is also fitted. Scores are measured on the reserved holdout split.

The mean-prediction baseline has holdout MSE 12.9872 for var1 and 46.1607 for var2. Both selected polynomial models substantially reduce this error. Larger degrees were evaluated but did not improve cross-validation MSE beyond the selected configurations.

Deliverables

Two prediction CSVs, saved model parameters, training and inference scripts, numerical verification, and recorded validation results accompany this report. The repository link appears on page 4.

Model selection

Polynomial features, regularization, and validation protocol

3. Polynomial ridge regression

The feature map contains every monomial whose total degree is at most d . For example, x_1 squared times x_2 is a degree-3 term. With p inputs, the number of nonconstant terms is $C(p + d, d) - 1$. This includes interactions between inputs, rather than only separate powers of each input.

```
Objective = sum of squared prediction errors
+ alpha * sum of squared coefficients
```

The intercept is unpenalized. Polynomial columns and the target are centered using only the current training partition. No column variance normalization is applied after expansion: ridge penalizes coefficients in the original monomial basis. Inputs are already bounded, limiting the magnitude of raw powers. The objective uses a sum of squared errors, not a mean, so alpha must be interpreted with that convention.

4. Search and split design

Unique input groups were shuffled with random seed 42. Twenty percent of groups were reserved for the holdout: 200 rows for var1 and 198 rows for var2. The remaining 800 and 802 development rows were divided into three grouped folds. The same folds were used for all candidates.

The search covered every degree from 1 to 10 for var1 and 1 to 20 for var2, with 23 logarithmically spaced alpha values from 10^{-8} to 10^3 at each degree. The configuration with the smallest mean fold MSE was selected. The holdout was not used to choose or revise these settings.

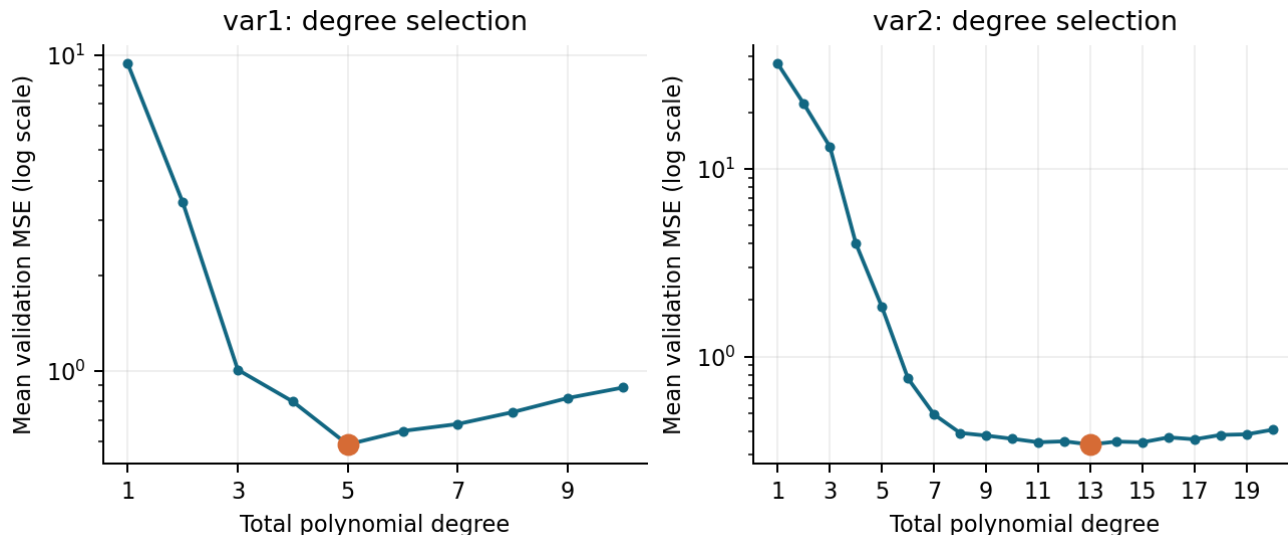


Figure 1. Best mean cross-validation MSE at each degree, after selecting alpha within that degree. Orange markers show the chosen degrees. The vertical axis is logarithmic.

Var1 reaches its minimum CV MSE of 0.584824 at degree 5; degree 10 gives 0.885626. Var2 reaches 0.339568 at degree 13; degree 20 gives 0.407304. The selected models therefore balance flexibility and regularization using observed validation performance.

Validation evidence

Predictions, residuals, and performance across the input range

After selection, each model was fitted on its full development partition and evaluated on its reserved holdout. MSE measures mean squared prediction error. R2 is 1 minus residual sum of squares divided by the total squared deviation of the evaluated targets from their own mean.

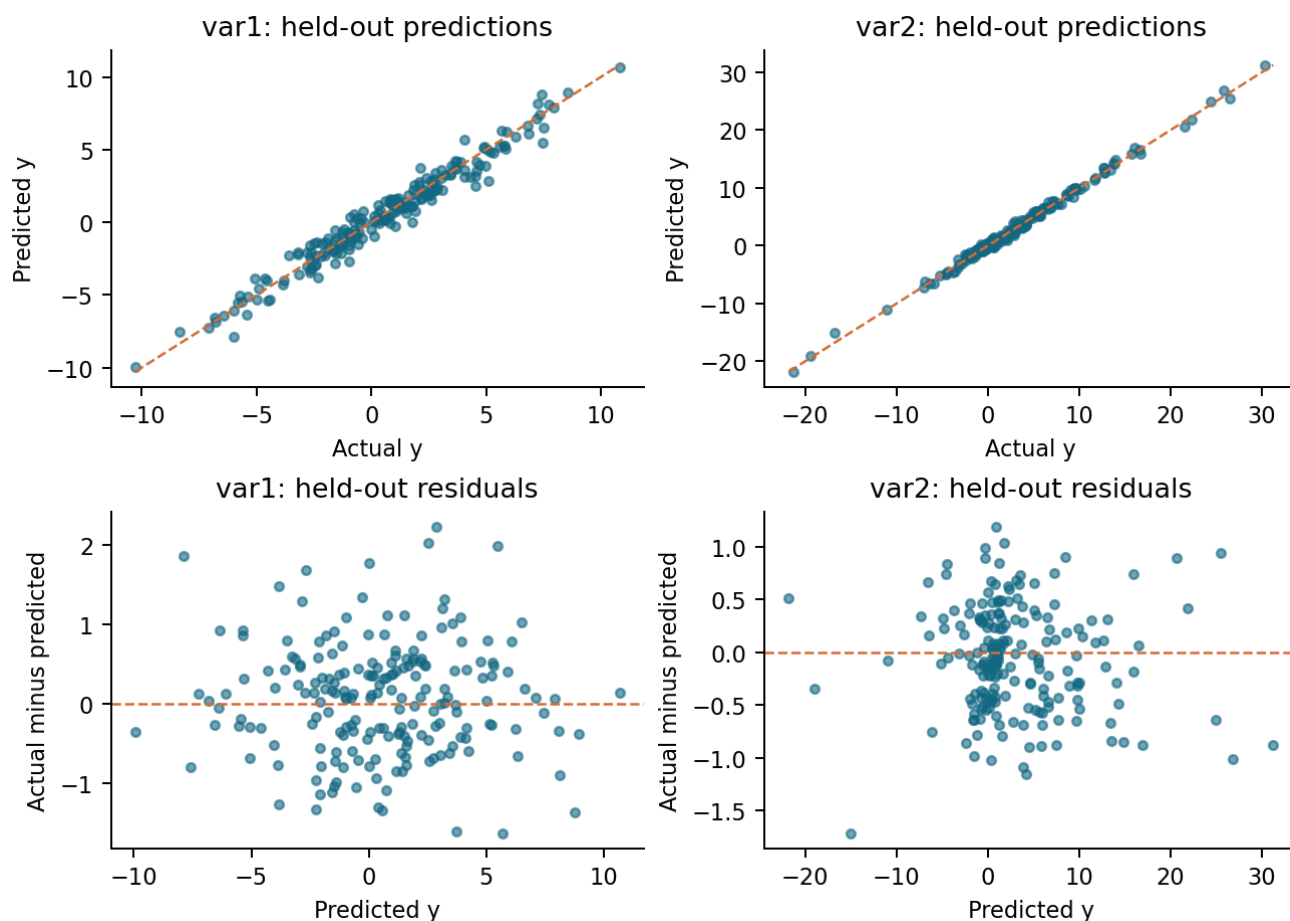


Figure 2. Top: actual versus predicted scores, with the perfect-prediction line. Bottom: residuals against predictions. Every point is a held-out row. The plots are diagnostics; no settings were changed after examining them.

5. Boundary diagnostics

Problem	Boundary rows: train / test	Boundary holdout MSE	Interior holdout MSE
var1	87.4% / 98.2%	0.517048	0.334074
var2	57.8% / 58.6%	0.286341	0.211126

A boundary row has at least one input with absolute value at least 0.999999. Var1 has more boundary rows in the test data than in training. Its random holdout may therefore not fully represent the test distribution. The boundary diagnostic is descriptive and was not used to retune the model.

Development-fit MSE is 0.192881 for var1 and 0.171147 for var2. The higher holdout errors show why training fit alone is insufficient for model selection.

Reproducibility and delivery

Final fitting, output verification, and remaining submission step

6. Final training and inference

After holdout evaluation, each selected configuration was refitted on all 1,000 labeled rows. The saved NPZ files contain the exponent matrix, coefficient vector, intercept, feature names, degree, and alpha. The inference script reconstructs the same polynomial features and computes one score per test row. No target clipping or postprocessing is applied.

NumPy and SciPy solve ridge regression using the smaller of the feature-space and sample-space Gram matrices. Validation reuses an eigendecomposition across alpha values; final fitting uses a positive-definite linear solve.

7. Reproduce the work

Use Python 3.12 with the dependencies in requirements.txt. Place the four personalized datasets and sample_submission.csv beside the scripts, then run:

```
python -m pip install -r requirements.txt
python train.py
python verify.py
python build_report.py
```

File	Purpose
polynomial.py / train.py	Polynomial terms, ridge solver, grouped selection, final fitting
predict.py	Load a saved model and generate predictions without retraining
verify.py	Independent numerical checks and submission validation
outputs/metrics.json	Selected configurations and recorded evaluation metrics
outputs/*_cv_results.csv	Every searched degree/alpha and its three fold errors
outputs/BT2024260_pred_var1.csv outputs/BT2024260_pred_var2.csv	Final test predictions: one y column and 1,000 rows per file

8. Verification and limitations

Both primal and dual ridge implementations passed comparison against an independent augmented least-squares solution. Checks also passed for known polynomial terms, group separation, saved-model predictions, finite output values, CSV column names, row counts, row order, and numerical CSV round-trip agreement. The original datasets were not modified.

The holdout is only one split. Predictions may be less reliable in sparsely sampled regions, particularly under the observed var1 input shift.

Repository

https://github.com/Lohith600/ML_ASSIGNMNET